Develop a conceptual data model reflecting the following requirements:

A. Identify the main entities.

   Outlet, Vehicle, Client, Hire Agreement, Staff

B. Identify the main relationships between the entities identified in "a".

   - Outlet has Vehicle
   - Outlet employs Staff
   - Client signs Hire Agreement
   - Vehicle included in Hire Agreement

C. Determine the multiplicity constraints for each relationship identified in "b".

   - Outlet has Vehicle: 1:* *(Assumption: each vehicle is associated with only one outlet.)*
   - Outlet employs Staff: 1:*
   - Client signs Hire Agreement: 1:* *(Assumption: only 1 client can sign a Hire Agreement.)*
   - Vehicle included in Hire Agreement: 1:* *(Assumption: only 1 vehicle is included in each Hire Agreement.)*

D. Identify attributes and associate them with entities or relationships.

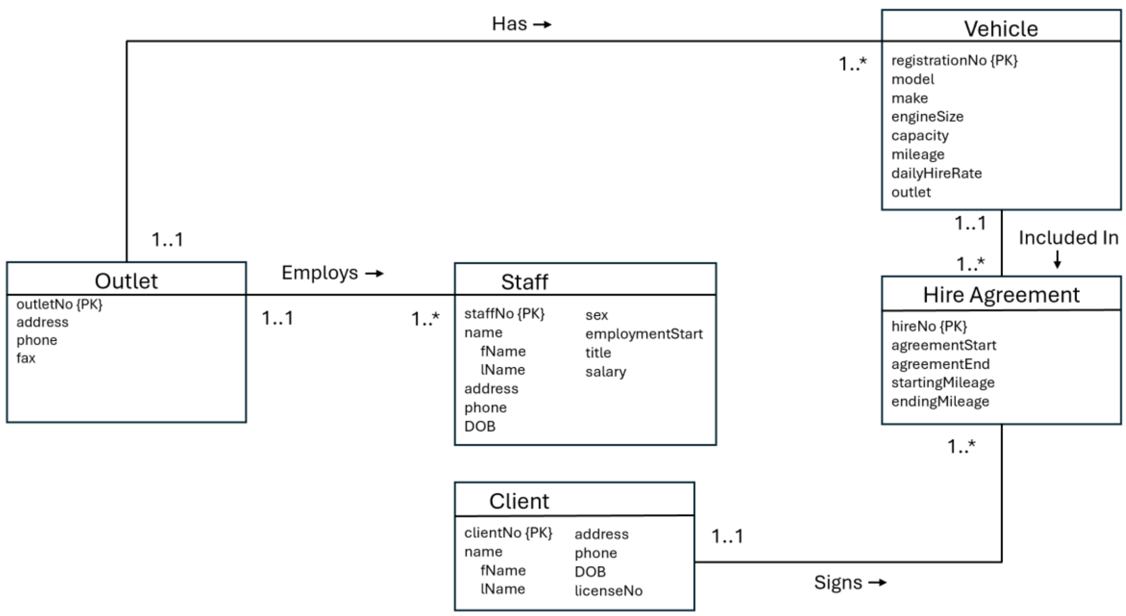
   - Outlet attributes: outletNo, address, phone, fax
   - Vehicle attributes: registrationNo, model, make, engineSize, capacity, mileage, dailyHireRate, outlet
   - Hire Agreement attributes: hireNo, clientNo *(foreign key so will be left out of the ER diagram at this stage)*, registrationNo *(foreign key so will be left out of the ER diagram at this stage)*, agreementStart, agreementEnd, startingMileage, endingMileage *(Note: client name, address, and phone number and vehicle model and make, though included in the hire agreements, should not be included in this entity, as in the future these can be identified via foreign keys to other tables and would need to be removed from the Hire Agreement table as part of the data normalization process.)*
   - Client attributes: clientNo, name (composite attribute comprised of fName and lName), address, phone, DOB, licenseNo
   - Staff: staffNo, name (composite attribute comprised of fName and lName), address, phone, DOB, sex, employmentStart, title, salary

E. Determine candidate and primary key attributes for each (strong) entity.

- Outlet:
    A. Candiate keys: outletNo, address *(Assumption: No two outlets exist at the same address.)*, phoneNo *(Assumption: Each outlet has a unique phone number.)*, faxNo *(Assumption: Each outlet has a unique fax number.)*
    B. Primary key: outletNo – *This is the option that is least likely to change*
- Vehicle:
    A. Candidate keys: registrationNo
    B. Primary key: registrationNo – *Only unique option*
- Hire Agreement:
    A. Candidate keys: hireNo; composite key of clientNo, registrationNo, agreementStart; composite key of registrationNo, agreementStart *(Assumption: Vehicles are only engaged in one Hire Agreement at a time, but clients can sign more than one Hire Agreement for the same time period, as in the case of signing on behalf of oneself as well as on behalf of a different driver/dependent minor.)*
    B. Primary key: hireNo – *Simplest*
- Client:
    A. Candidate keys: clientNo, licenseNo
    B. Primary key: clientNo – *More in control of this*
- Staff:
    A. Candidate keys: staffNo
    B. Primary key: staffNo – *Only unique option*

F. Generate the E-R diagram for the conceptual level (no FKs as attributes).

Notes: A report must be created including detailed documentation of each of the steps addressing all the required items. ER diagrams for the conceptual model must be included in the report. All assumptions made in the design must be clearly stated.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